Santiago Osorio Jurado

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RESEARCH INTERESTS

Neuromodulation for psychiatric disorders and for motor recovery after stroke and spinal cord injury, with the goal of therapies patients can use at home to restore function rather than only manage symptoms.

EDUCATION

Bachelor of Engineering (B.E.), Biomedical Engineering Aug. 2022 – May 2026
The City College of New York, CUNY | Grove School of Engineering New York, NY

- Magna Cum Laude, GPA 3.56/4.00; Dean's List; Tau Beta Pi (top eighth of juniors, 2025).

PUBLICATIONS

Osorio Jurado, S., Skorpil, M., Svenningsson, P., Moreno, R., & Olsson, C. (2026). Anisotropic conductivity modeling for tDCS in Parkinson's disease using multidimensional diffusion MRI. *Frontiers in Human Neuroscience*, 20, 1972676. https://doi.org/10.3389/fnhum.2026.1972676

Donnery, K., FallahRad, M., **Osorio, S.**, Belali Koochesfahani, M., Khadka, N., Saw, M., Mo, J., Babaev, B., Bhuiyan, R., Elwassif, J. M., & Bikson, M. (2026). High-capacity transcranial direct current stimulation (HC-tDCS). *Brain Stimulation*, Article 103233. https://doi.org/10.1016/j.brs.2026.103233

Osorio Jurado, S., Wang, B., & Bikson, M. (in preparation). Transverse polarization in neuromodulation.

Huh, Y. E.†, Oh, S.†, Ahn, J., You, S.-H., **Osorio Jurado, S.**, Belali Koochesfahani, M., Bikson, M., Park, H., Youn, J., & Cho, J. W. (in preparation). Home-based cerebellar transcranial direct current stimulation for cerebellar variant of multiple system atrophy: A randomized controlled trial. †Equal contribution.

RESEARCH EXPERIENCE

CCNY Neural Engineering Group | PI: Prof. Marom Bikson New York, NY
Research Engineer Aug. 2026 – Present

- Scaling up electrode manufacturing for two trials at UT Southwestern (late-life depression and addiction).
- Developed a single-use carbon version of the electrodes and tested it against silver at 6 mA in 10 volunteers (IRB-approved); tolerability was comparable, qualifying it for these trials.
- Building individualized cerebellar tDCS models (SimNIBS, Simpleware, COMSOL) for a home-based randomized trial in multiple system atrophy (CHA Bundang and Samsung Medical Center), testing whether each patient's cerebellar electric field predicts clinical improvement.

Undergraduate Researcher Jan. 2025 – May 2026

- Engineered multilayer Zn/AgCl redox electrodes with a hydrogel interface (≤ 600 Ω·cm) enabling safe stimulation at 6 mA, triple the conventional 2 mA dose; verified stability via EIS, overpotential sweeps, and SEM.
- Ran a single-blind, IRB-approved study in 10 healthy volunteers at 1–6 mA (VAS pain, PANAS, Draize erythema, thermal imaging); pain at 6 mA was comparable to 2 mA with standard sponge electrodes.
- Developed the MRI-derived COMSOL field models (2.25–3.26 V/m peak electric fields at 6 mA) and the custom PCB stimulator hardware; quantified dose-response with linear mixed-effects models in Python.

Visiting Researcher (IRES Fellowship) Jun. – Aug. 2026
KTH Royal Institute of Technology | PI: Prof. Rodrigo Moreno Stockholm, Sweden

- Built a subject-specific finite-element pipeline (SimNIBS) deriving anisotropic conductivity tensors from multidimensional diffusion MRI (QTI); to our knowledge the first human tDCS field model to use MD-dMRI-derived conductivity.
- Compared isotropic, single-shell DTI, and QTI conductivity across four montages in 29 participants (12 Parkinson's, 17 controls); the anisotropic models differed by 6.6% or less in electric field, so anatomy dominated.
- Proposed the study and carried it from design to journal submission in one summer (first-author paper, Frontiers in Human Neuroscience); benchmarked electric field predictions against MR elastography stiffness.

Machine Learning Research Intern (SUPERB Fellowship) Jun. – Aug. 2025
Computational Imaging Lab, UC Berkeley | PI: Prof. Laura Waller Berkeley, CA

- Benchmarked SID (a U-Net) against SNR-Aware (a CNN-transformer hybrid) for denoising low-SNR darkfield microscopy; identified SID as stronger by PSNR, SSIM, and error maps on simulated and experimental targets.
- Trained both models from scratch on identical splits, using a 768-pair simulated dataset built from high-SNR cells (gamma correction, Poisson-Gaussian noise) and 720 experimental pairs acquired on USAF phase targets.

Medical Imaging Research Intern (IRES Fellowship) May – Sep. 2024
KTH Royal Institute of Technology | PI: Prof. Mats Persson Stockholm, Sweden

- Identified the settings that make iodine contrast easiest to see on a photon-counting CT (90 kVp, 40 keV reconstruction), testing 70–140 kVp at two dose levels on a Siemens NAEOTOM Alpha.
- Wrote a settings guide for radiologists new to photon-counting CT, based on an automated Python analysis.

SELECTED PRESENTATIONS & POSTERS

Osorio Jurado, S., Moreno, R., & Olsson, C. (2026, August). Anisotropic conductivity modeling for tDCS in Parkinson's disease using multidimensional diffusion MRI [Poster]. NYC Neuromodulation Conference, New York, NY.

Osorio Jurado, S., Moreno, R., & Olsson, C. (2026). Anisotropic conductivity modeling for tDCS in Parkinson's disease using multidimensional diffusion MRI [Poster]. Urban University Conference (UUC2026), KTH, Stockholm.

Osorio, S., Donnery, K., FallahRad, M., Belali Koochesfahani, M., Elwassif, J. M., Saw, M., Babaev, B., Bhuiyan, R., Khadka, N., & Bikson, M. (2025, November). High-capacity transcranial direct current stimulation (HC-tDCS) [Poster]. Annual Biomedical Research Conference for Minoritized Scientists (ABRCMS), San Antonio, TX.

Osorio, S. (2025, August). Deep-learning denoising for extreme short-exposure darkfield microscopy [Talk]. SUPERB Summer Research Symposium, UC Berkeley.

Osorio, S. (2024). Optimal tube-voltage selection in photon-counting CT for iodine contrast imaging [Poster]. IRES Research Symposium, KTH / CCNY.

HONORS, AWARDS & FELLOWSHIPS

NIH U-RISE Scholar (NIGMS T34), National Institutes of Health	2025 – 2026
Rukin Award for Academic and Professional Perseverance in BME, CCNY	2026
Wallace H. Coulter Award for Academic Service in BME, CCNY	2025, 2026
BMES Student Chapter Executive Board Award, CCNY	2025, 2026
Endowed Scholarship to Support the Pursuit of Engineering Careers, CCNY	2025
America Needs You Fellowship	2023 – 2024

SELECTED PROJECTS

Thermal Stimulator for Automated Rodent Pain Testing (Senior Design Capstone) Aug. 2025 – May 2026
CCNY Biomedical Engineering | Sponsor: Dr. Justin Burdge, Tactorum New York, NY

- Wrote the control software that holds temperature within ±0.5 °C and tracks a 1 °C/s ramp through a four-stage cycle (cool, preheat, ramp, hold), adjustable over a serial connection while running.
- Fixed a ramp that consistently fell short by tracing it to a control-logic reset at the constant-to-ramp handoff and retuning the PID controller.
- Built and validated the Arduino power electronics as a plug-in module for Tactorum's ARM rodent-testing platform; with the team's Peltier cooling stage, cool-down between tests fell from 417 to 66 seconds.

LEADERSHIP, MENTORING & SERVICE

Chapter Officer (Secretary), Biomedical Engineering Society (BMES) 2024 – 2026
CCNY Chapter New York, NY

- Co-organized the Biodesign Challenge for two years, a medical-device prototyping competition (106 students; $7,000 in prizes); secured sponsor funding, recruited judges, and taught soldering, CAD, and pitching workshops.
- Led ResearchConnect, which helps students find labs and approach professors, placing eight undergraduates in their first lab positions.

Chapter Officer (Secretary), Tau Beta Pi Engineering Honor Society 2025 – 2026
CCNY Chapter (New York Eta) New York, NY

TECHNICAL SKILLS

Programming & computing: Python (NumPy, SciPy, pandas, PyTorch, Matplotlib), MATLAB, C/C++ (Arduino), Bash; Git; LaTeX; HPC / SLURM clusters.
Neural & biomedical modeling: finite-element field modeling (COMSOL, SimNIBS); MRI processing (FreeSurfer, FSL, ANTs, SAMSEG); diffusion MRI (DTI, QTI / b-tensor encoding); anisotropic conductivity estimation; statistical modeling (mixed-effects, FDR).
Machine learning & signal processing: deep learning for image restoration and denoising in PyTorch (U-Net, CNN-transformer hybrids); synthetic data generation; PSNR/SSIM evaluation.
Electronics & devices: PCB design and stimulator electronics; electrode fabrication; electrochemical impedance spectroscopy (EIS); SEM; embedded systems (Arduino, MOSFET drivers); closed-loop PID control; 3D CAD (SolidWorks).
Experimental & translational: IRB human-subjects protocols; single-blind randomized crossover design; tolerability assays (VAS, Draize, PANAS); photon-counting CT.
Languages: Spanish (native); English (fluent, professional).